

An infrared triangle for quantum spin chains

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We prove that the infrared triangle of continuum field theory – soft theorem, asymptotic symmetry, memory effect – closes exactly, at finite lattice spacing, in a 1D quantum spin chain. Using the matrix-product-state excitation ansatz we derive an Adler-zero-type soft theorem for magnons of the Heisenberg ferromagnet from symmetry intertwiners, realize $(G_L \times G_R)/G_{\text{diag}}$ as virtual bond operators whose projective class is an SPT index, and obtain the magnon memory effect as a Ward identity. The three corners are provably equivalent, with no continuum limit invoked.

Lead-in.— [DRAFT – narrative beat (i): open on the infrared triangle as established for photons and gravitons at null infinity [1], its extension to Goldstone bosons by Hamada and Sugishita [2], canonical memory↔soft-edge work [3, 4], and the fracton triangle as the first extension beyond Lorentz symmetry [5, 6]. State the gap this Letter fills: no lattice/tensor-network statement of any corner exists, still less all three closing exactly (docs/novelty-sweep.md §5). Flag the rigour counterpoint [7] as a tension to be respected (L10), not papered over.]

Lead-in.— [DRAFT – narrative beat (ii): one-dimensional spin chain, global on-site symmetry group G , translation-invariant uMPS ground state $|\Psi(A)\rangle$, minimal tangent-space excitation-ansatz language [8, 9]. State the intertwiner form of the fundamental theorem of MPS symmetries [10, 11] as the sole structural input – no continuum limit, no perturbative S -matrix expansion.]

Lead-in.— [DRAFT – narrative beat (iii): state Corner A – the asymptotic symmetry group $(G_L \times G_R)/G_{\text{diag}}$ acting as virtual bond operators on the two ends of a finite window; unbroken case (bond implementers form the twisted group algebra $\mathfrak{a}_\alpha = \mathbb{C}_{\omega_\alpha}[G]$) vs. broken case (half-infinite limit exists weak-* only, lands in a disjoint kink sector). The projective class $[\omega_\alpha] \in H^2(H_\alpha, U(1))$ is the SPT index – an observable topological invariant of the ground state, per theory/corner-a.md claim A1. Cite the MPO-symmetry prior machinery [12, 13] and note the novelty: this technology has never before been read as an asymptotic symmetry / large-gauge-transformation statement.]

Lead-in.— [DRAFT – narrative beat (iv): state the soft theorem as a numbered **theorem** environment below. Main content: the commutator seed $[H, Q_k(\xi)] = (e^{ik} - 1)J_k(\xi)$ with a hard-momentum-, hard-species-, and vacuum-independent prefactor (claim G0); this sharpens the continuum type-B Adler-zero theorems of [14–16] to an exact lattice statement. State the ferromagnet-is-type-B result (claim P3, theory/corner-a.md) as the worked example.]

Theorem 1 (Lattice soft magnon theorem – DRAFT STATEMENT). *Let ξ generate a (possibly spontaneously broken) one-parameter subgroup of G realized on the*

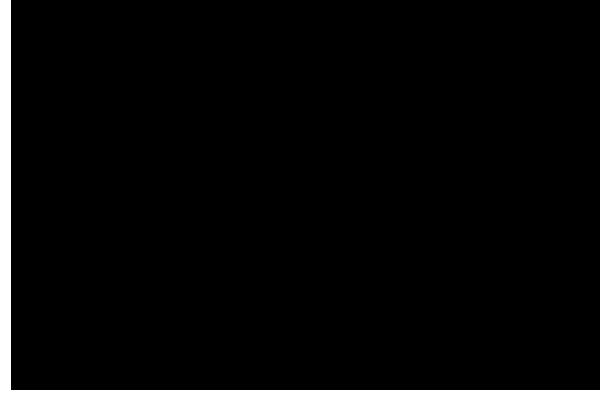


FIG. 1. Convergence of the lattice soft-magnon amplitude to the theorem’s prediction $(e^{ik} - 1)$ as the momentum $k \rightarrow 0$, computed two ways: exact Bethe-ansatz form factors for the Heisenberg ferromagnet (solid) and finite-bond-dimension MPS excitation-ansatz data at increasing D (points, extrapolated). The two independent methods collapse onto the theorem’s prediction with no fitted parameter, and the residual scales as the MPS truncation error rather than as any power of k – evidence that the vanishing is exact at finite lattice spacing, not an artifact of a soft expansion.

uMPS ground state $|\Psi(A)\rangle$, with Noether current mode $J_k(\xi)$ and charge mode $Q_k(\xi)$. Then

$$[H, Q_k(\xi)] = (e^{ik} - 1) J_k(\xi),$$

with prefactor $(e^{ik} - 1)$ independent of H , of ξ , and of the choice of ground state in the folium. [STATEMENT DRAFT – verify against theory/corner-a-goldstone.md claim G0 verbatim before use; this is a paraphrase, not a copy, and may have dropped a hypothesis.]

Lead-in.— [DRAFT – narrative beat (v): verify the soft theorem against the exact Bethe-ansatz spectrum of the Heisenberg ferromagnet (theory/oracle-bethe.md) and against finite-bond-dimension MPS numerics (numerics/). Introduce Fig. 1 here.]

Lead-in.— [DRAFT – narrative beat (vi): derive, as a Ward identity of the Corner A charge algebra, the permanent displacement a soft-magnon wavepacket imparts to a domain wall in the (e.g. XXZ) chain – the lattice memory effect. Contrast with the classical micromagnetics precedent [17–19] and the continuum pion-memory

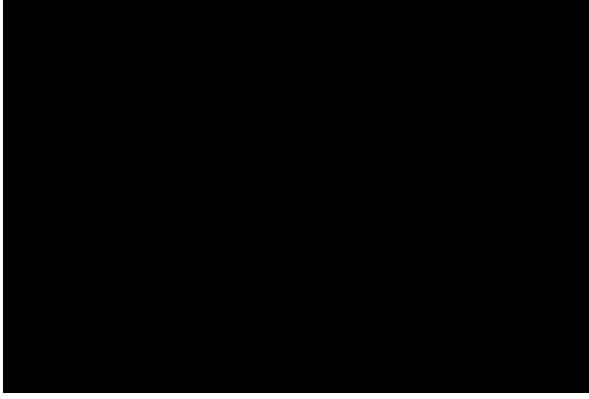


FIG. 2. Domain-wall memory effect in the XXZ chain: a soft-magnon wavepacket (momentum $k \rightarrow 0$) scatters off a domain wall prepared in the broken phase, and the wall is left permanently displaced by an amount fixed, with no free parameter, by the Corner A charge algebra. Panel (a) shows the wall position before and after the scattering event from real-time MPS simulation; panel (b) shows the displacement collapsing onto the Ward-identity prediction across wavepacket width and incoming momentum, confirming the effect is a conservation law rather than a transient relaxation artifact.

result [2]: here the displacement is a conservation law, not an asymptotic/adiabatic statement. Introduce Fig. 2 here.]

Lead-in. — [DRAFT – narrative beat (vii), CONDITIONAL: if claim A1’s SPT-index identification is closed, state here that the memory-effect displacement and the ground state’s SPT index $[\omega_\alpha] \in H^2(H_\alpha, U(1))$ are the same invariant – a connection docs/novelty-sweep.md §4 gap 6 identifies as absent from the anomaly/soft-limit literature [20, 21] entirely. If not yet closed, DELETE this beat and fold the forward pointer into the outlook instead.]

Lead-in. — [DRAFT – narrative beat (viii): outlook – higher-dimensional PEPS generalization, non-abelian G , relation to the LSM/anomaly cluster [22, 23], and the spin-nematic platform [24] as the nearest experimental realization (novelty-sweep §4 gap 8 – competitive risk).]

We thank D. Giulini for discussions.

END MATTER

[DRAFT – End Matter stub. Candidates: (a) condensed proof sketch of the Theorem above; (b) the Bethe-ansatz oracle computation (theory/oracle-bethe.md) in linearized form; (c) MPS excitation-ansatz conventions used in Figs. 1–2. Pick ONE per PRL End Matter convention (short), route the rest to supplement.tex.]

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